Progress update

Making quantum circuits shorter, and proving how short they can get

Exact ground truth for local rewriting, and a general hardware-aware optimiser

Arnav Kapoor

IISER Bhopal

Building on B. Rosenhahn, T. J. Osborne, C. Hirche, *Optimization driven quantum circuit reduction*, New J. Phys. **27**, 104509 (2025)

The story in one slide

The starting point

- A recent paper shortens circuits by swapping small pieces for shorter ones from a precomputed table.
- It is compared only with Qiskit, on random circuits.

What we found

- Every gate it uses is **Clifford**, so the true optimum is computable.
- Against it, the method is **3–4.5**$\times$ too long, and we can say why.

The story in one slide

The starting point

- A recent paper shortens circuits by swapping small pieces for shorter ones from a precomputed table.
- It is compared only with Qiskit, on random circuits.

What we found

- Every gate it uses is **Clifford**, so the true optimum is computable.
- Against it, the method is **3–4.5**$\times$ too long, and we can say why.

What we built

- A **general optimiser** for real native gates, connectivity and calibrated noise, exact including phase.

Where it stands

- Proven optimum on **35/40** of the paper's 3-qubit circuits.
- After Qiskit L3 on 58 algorithms it never adds a 2q gate and removes **13 %** of gates (Heron).
- Up to 95 qubits it removes **19 %** on top of Qiskit L3, every output certified.
- On the standard Nam set, VOQC then ours is the best of six tools ($-25\,\%$).

Part I

The method we start from

Replace short windows of a circuit with shorter equivalents from a precomputed database.

- On today's hardware every gate costs fidelity. Roughly,

$$\Pr[\text{success}] \approx \prod_{\text{gates } g} \big(1 - e(g)\big),$$

  the **ESP**. Two-qubit errors are 10–100× single-qubit ones.
- Compiling to a device's **native gates** leaves redundancy.
- Goal: an *equivalent* circuit with fewer, or cheaper, native gates.

Figure from arXiv:2502.14715, Fig. 1

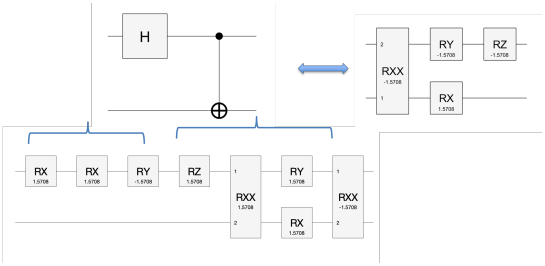
Why shorten circuits at all?

- On today's hardware every gate costs fidelity. Roughly,

$$\Pr[\text{success}] \approx \prod_{\text{gates } g} \big(1 - e(g)\big),$$

  the **ESP**. Two-qubit errors are 10–100× single-qubit ones.

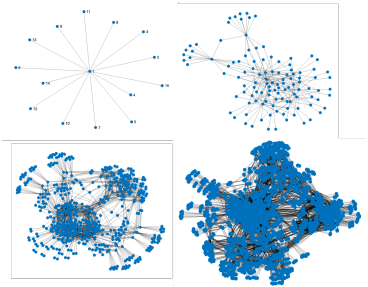
- Compiling to a device's **native gates** leaves redundancy.
- Goal: an *equivalent* circuit with fewer, or cheaper, native gates.



H + CNOT compiled to ion-trap gates (8 gates) and the shortest equivalent (4 gates).

Figure from arXiv:2502.14715, Fig. 1

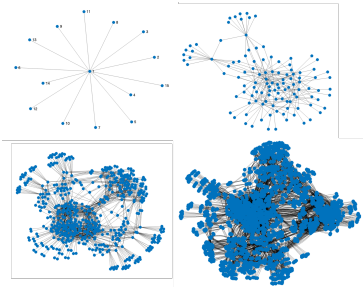
The method: a database and a sliding window



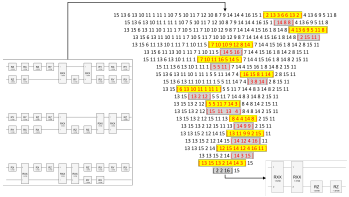
Compute graph. Nodes are circuits (as matrices), edges add one gate. Everything within radius r of the identity is stored with its shortest form. The paper uses $r = 5$–6.

Figures from arXiv:2502.14715, Figs. 2 and 3

The method: a database and a sliding window



Compute graph. Nodes are circuits (as matrices), edges add one gate. Everything within radius r of the identity is stored with its shortest form. The paper uses $r = 5$–6.



Window. Take 2–8 gates on ≤ 3 wires, look them up, substitute a shorter word, shuffle commuting gates, repeat.

Takeaway. Two knobs decide everything later: the window length w and the database radius r.

Figures from arXiv:2502.14715, Figs. 2 and 3

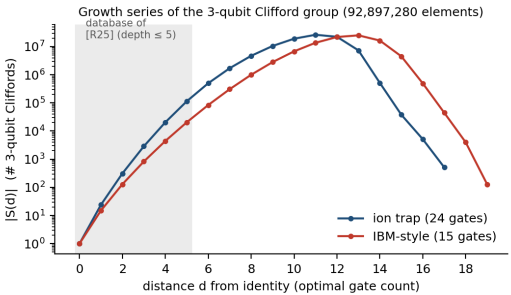
Part II

Exact ground truth

Every gate the paper uses is Clifford, so the search space is a finite group we can enumerate completely.

Every gate in the paper is Clifford

- $\pm\frac{\pi}{2}$ rotations, $R_{XX}(\pm\frac{\pi}{2})$ and CZ map Paulis to Paulis.
- So every circuit is an element of the finite **Clifford group**: 24 elements for 1 qubit, 11,520 for 2, **92,897,280** for 3.
- We enumerated all of them for 3 qubits and stored the exact shortest length of each (25 min, 6 GB).

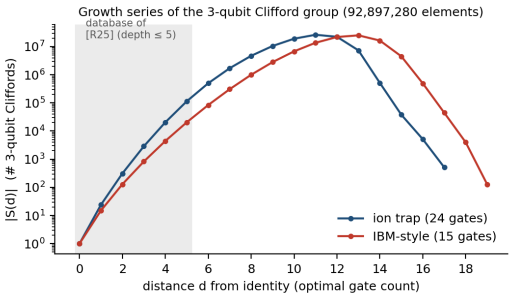


Elements by optimal length (3 qubits). The paper's depth-5 database (shaded) covers 0.15 % of the group.

Every gate in the paper is Clifford

- $\pm\frac{\pi}{2}$ rotations, $R_{XX}(\pm\frac{\pi}{2})$ and CZ map Paulis to Paulis.
- So every circuit is an element of the finite **Clifford group**: 24 elements for 1 qubit, 11,520 for 2, **92,897,280** for 3.
- We enumerated all of them for 3 qubits and stored the exact shortest length of each (25 min, 6 GB).

Takeaway. Optimality is no longer vague. Each test circuit has a proven shortest length.



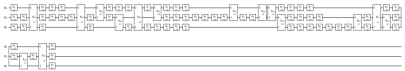
Elements by optimal length (3 qubits). The paper's depth-5 database (shaded) covers 0.15 % of the group.

How far from optimal is the paper's method?

3 qubits, 300 random gates, 20 circuits per row

	optimum	paper's method	Qiskit L3
ion trap	**10.4**	47.1	127.2
IBM-style	**12.2**	36.8	104.0

(a) input: random 3-qubit word over the ion alphabet, 80 gates



(b) original method (database radius 5): 24 gates



(c) proven optimum (exact Cayley table): 11 gates



A typical circuit: 80 gates in, the paper's method stops at 24, the proven optimum is 11.

How far from optimal is the paper's method?

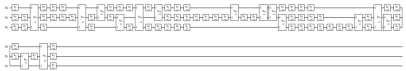
3 qubits, 300 random gates, 20 circuits per row

	optimum	paper's method	Qiskit L3
ion trap	**10.4**	47.1	127.2
IBM-style	**12.2**	36.8	104.0

The gap

The method's output is **4.5**× (ion) and **3.0**× (IBM-style) the optimum, on its own benchmark. Our port matches its published number (47.1 vs. 45), so this is the method, not our code.

(a) input: random 3-qubit word over the ion alphabet, 80 gates



(b) original method (database radius 5): 24 gates

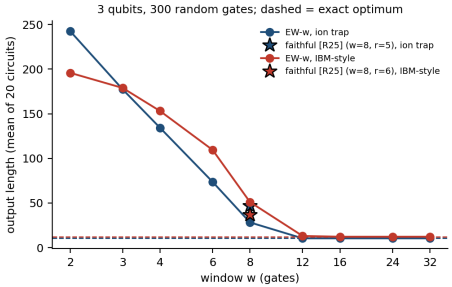


(c) proven optimum (exact Cayley table): 11 gates



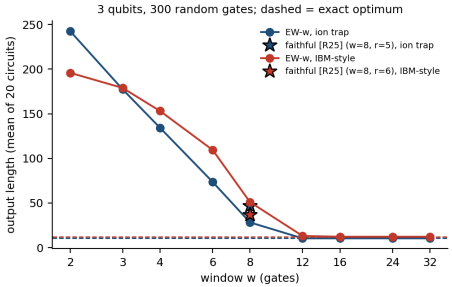
A typical circuit: 80 gates in, the paper's method stops at 24, the proven optimum is 11.

Why it falls short: the database is too small

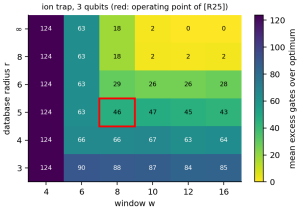


With exact lookups, a window of $w \geq 12$ (ion) or 16 (IBM-style) reaches the proven optimum on all 40 circuits.

Why it falls short: the database is too small



With exact lookups, a window of $w \geq 12$ (ion) or 16 (IBM-style) reaches the proven optimum on all 40 circuits.



Mean excess over the optimum by window w and radius r.

The paper's $(8, 5)$ in red.

- Larger window at $r = 5$: excess 46 $\rightarrow$ 43.
- Larger database at $w = 8$: 46 $\rightarrow$ 18.
- $w \geq 12$ and $r \geq 8$: excess ≤ 2.

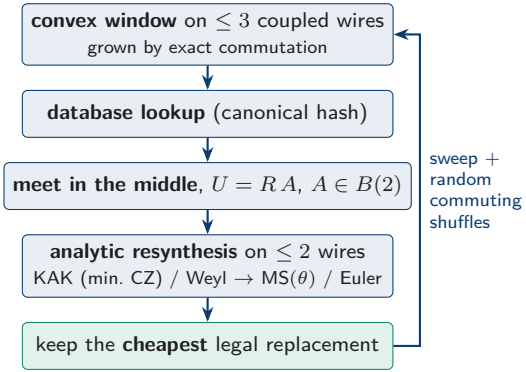
Takeaway. The database radius is the binding constraint. Our theorem predicts that both knobs must be large enough.

Part III

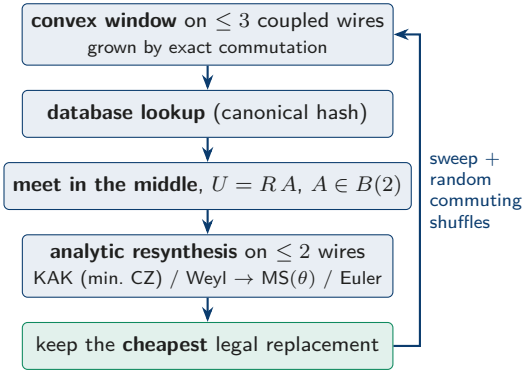
A general optimiser

Drop the Clifford crutch and add continuous angles, connectivity, calibrated noise, and exactness including phase.

The engine


convex window on ≤ 3 coupled wires
grown by exact commutation
database lookup (canonical hash)
meet in the middle, $U = R\,A$, $A \in B(2)$
analytic resynthesis on ≤ 2 wires
KAK (min. CZ) / Weyl $\rightarrow$ MS(θ) / Euler
keep the **cheapest** legal replacement
sweep + random commuting shuffles

The engine



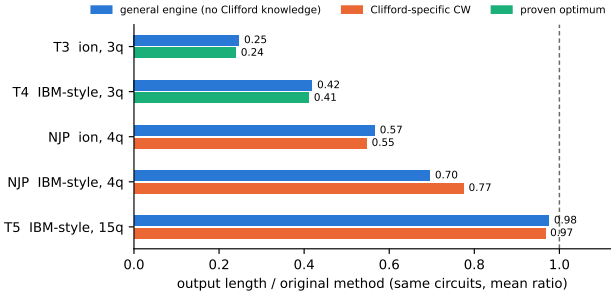
- **Cost** $c_{\mathrm{cal}}(g) = -\log(1 - e(g))$ from calibration, so minimising cost maximises ESP.
- **Exact including global phase**: phases are tracked and repaired once at the end.
- **Never worse** than its input, by construction.
- **Any gate set**: databases of 0.3–31 M classes for IBM Heron and trapped ions.
- Runs *after* any compiler, as a post-pass.

Part IV

Results

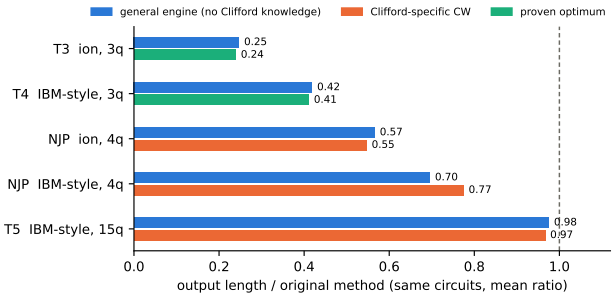
The paper's benchmarks against ground truth, 60 algorithm circuits on hardware models, and circuits up to 100 qubits.

Result 1. The paper's benchmarks, without Clifford knowledge



Output length relative to the paper's method, same circuits and seeds.

Result 1. The paper's benchmarks, without Clifford knowledge



Output length relative to the paper's method, same circuits and seeds.

- **3 qubits.** Proven optimum on **35/40**, mean 10.8 vs. optimum 10.4.
- **4 qubits.** On par with the Clifford-specific optimiser.
- **15 qubits.** All methods within 4 % (random circuits leave little local structure).
- **Clifford+T, 2 qubits.** Optimal on 99/99 random and 41/41 planted circuits.

Takeaway. Knowing the group is not what matters. Large databases and convex windows are.

Result 2. 60 algorithm circuits, relative to Qiskit level 3

	IBM Heron (58 circuits)			**trapped ion** (58 circuits)	
pipeline	2q gates	all gates	ESP	2q gates	all gates
TKET	×1.02	×1.71	×1.16	×1.27	×1.79
ours, exact, from Qiskit L1	×1.00	**×0.84**	**×1.21**	**×0.95**	**×0.76**
ours, exact, from Qiskit L3	**×0.98**	**×0.87**	×1.03	×1.00	×1.00
BQSKit L3 on L1 (approximate)	**×0.85**	×1.20	**×1.22**	**×0.94**	**×0.71**
ours, exact, on BQSKit L3	**×0.85**	×1.01	**×1.25**	**×0.94**	**×0.70**

Seed 0. Over 5 seeds the gate ratios move by at most 0.5 % and ESP from L1 is $\times 1.10 \pm 0.11$. Nam/VOQC, QASMBench, QFT, QAOA, Trotter, EfficientSU2, Steane, random, 2–12 qubits.

Result 2. 60 algorithm circuits, relative to Qiskit level 3

	IBM Heron (58 circuits)			**trapped ion** (58 circuits)	
pipeline	2q gates	all gates	ESP	2q gates	all gates
TKET	×1.02	×1.71	×1.16	×1.27	×1.79
ours, exact, from Qiskit L1	×1.00	**×0.84**	**×1.21**	**×0.95**	**×0.76**
ours, exact, from Qiskit L3	**×0.98**	**×0.87**	×1.03	×1.00	×1.00
BQSKit L3 on L1 (approximate)	**×0.85**	×1.20	**×1.22**	**×0.94**	**×0.71**
ours, exact, on BQSKit L3	**×0.85**	×1.01	**×1.25**	**×0.94**	**×0.70**

- **Stacked on L3** ours never adds a 2q gate and never lowers ESP.
- **BQSKit** finds better 2q structure but is approximate and leaves many 1q gates. Ours on top keeps its 2q savings and removes **16 %** more gates on Heron, exactly.

Seed 0. Over 5 seeds the gate ratios move by at most 0.5 % and ESP from L1 is $\times 1.10 \pm 0.11$. Nam/VOQC, QASMBench, QFT, QAOA, Trotter, EfficientSU2, Steane, random, 2–12 qubits.

Which objective? Let noisy simulation decide

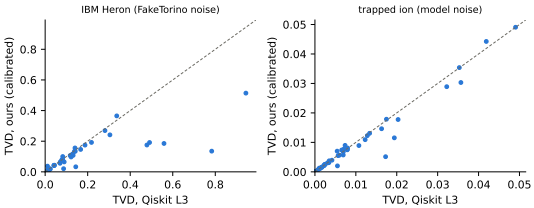
pipeline / objective	mean TVD (49 circuits) Heron	ion
Qiskit L3	0.137	0.0091
ours, total count	0.096	0.0092
ours, 2q-weighted	**0.088**	**0.0084**
ours, calibrated	**0.088**	**0.0084**

Exact density-matrix simulation with FakeTorino errors (Heron) or a depolarising model (ion).

Which objective? Let noisy simulation decide

pipeline / objective	mean TVD (49 circuits) Heron	ion
Qiskit L3	0.137	0.0091
ours, total count	0.096	0.0092
ours, 2q-weighted	**0.088**	**0.0084**
ours, calibrated	**0.088**	**0.0084**

Exact density-matrix simulation with FakeTorino errors (Heron) or a depolarising model (ion).



Calibrated beats Qiskit L3 on **39/49** Heron circuits.

Takeaway. Counting 2q gates with weight 10 is almost as good as full calibration, a useful proxy when no calibration exists.

Result 3. Large circuits, 14–100 qubits

Relative to Qiskit L3

	Heron (22)			**ion** (25)	
	2q	all	ESP*	2q	all
Qiskit L1	×1.22	×1.13	×0.24	×1.40	×1.13
TKET	×1.07	×1.96	×0.27	×1.39	×1.87
ours from L1	×1.05	**×0.78**	×0.43	**×0.96**	**×0.68**
ours from L3	**×0.98**	**×0.81**	**×1.12**	×1.00	×1.00

45 Nam/VOQC circuits, 88 of 90 runs done. pf circuits excluded (Qiskit L3 drops their 10^{-7} rad rotations). *Over the 13 Heron circuits whose ESP is at least 10^{-3}.

Result 3. Large circuits, 14–100 qubits

Relative to Qiskit L3

	Heron (22)			**ion** (25)	
	2q	all	ESP*	2q	all
Qiskit L1	×1.22	×1.13	×0.24	×1.40	×1.13
TKET	×1.07	×1.96	×0.27	×1.39	×1.87
ours from L1	×1.05	**×0.78**	×0.43	**×0.96**	**×0.68**
ours from L3	**×0.98**	**×0.81**	**×1.12**	×1.00	×1.00

45 Nam/VOQC circuits, 88 of 90 runs done. pf circuits excluded (Qiskit L3 drops their 10^{-7} rad rotations). *Over the 13 Heron circuits whose ESP is at least 10^{-3}.

- **Stacked on L3, Heron.** 19 % fewer gates, fewer CZ on 20/22 circuits, never more.
- **From L1.** Fewer gates than L3 everywhere, but lower ESP: at this scale Qiskit's L3 layout matters more than what we remove.
- **Cost.** Minutes per circuit, up to 6 h for the largest, single-threaded Python.
- **BQSKit** (14 circuits it finished, ≤36 qubits). Heron 2q ×0.91 vs. ours ×0.98, all gates ×1.35 vs. ×0.82.

Takeaway. Larger circuits leave more to remove, and the guarantees hold unchanged.

Result 4. Robust over seeds, and what each part contributes

5 transpiler seeds, Heron (mean ± sd vs. L3)

ours on L3, all gates	$\times 0.868 \pm 0.005$
ours on L3, 2q gates	$\times 0.975 \pm 0.004$
ours on L3, ESP	$\times 1.03$ (1.03–1.04)
ours on L1, ESP	$\times \mathbf{1.10 \pm 0.11}$

Gate counts are stable. ESP from L1 follows Qiskit's L1 layout, and seed 0 ($\times 1.21$) was the best case.

Result 4. Robust over seeds, and what each part contributes

5 transpiler seeds, Heron (mean ± sd vs. L3)

ours on L3, all gates	$\times 0.868 \pm 0.005$
ours on L3, 2q gates	$\times 0.975 \pm 0.004$
ours on L3, ESP	$\times 1.03$ (1.03–1.04)
ours on L1, ESP	$\times \mathbf{1.10 \pm 0.11}$

Gate counts are stable. ESP from L1 follows Qiskit's L1 layout, and seed 0 ($\times 1.21$) was the best case.

Ablation, gates removed on top of L3 (Heron)

full optimiser	**13.0 %**
contiguous windows	11.7 %
no resynthesis	8.2 %
window $w = 8$	7.9 %
database radius ≤ 3	6.2 %
no database	5.8 %
window $w = 4$	**1.3 %**

Takeaway. Window length and database depth do the work, the same two knobs as in Part II.

Result 5. Against VOQC and QUESO, on their own benchmark

Nam gate set $\{H, X, R_z, \mathrm{CNOT}\}$, 26 circuits

tool	gates	change	CNOT
input	10,481		4,612
TKET	11,762	$+12.2\,\%$	4,489
Qiskit L3	10,391	$-0.9\,\%$	4,511
QUESO (5 min)	9,765	$-6.8\,\%$	4,465
ours	9,480	$-9.6\,\%$	4,426
VOQC	8,417	$-19.7\,\%$	4,314
VOQC, then ours	**7,833**	$-25.3\,\%$	**4,007**

Same input files as the published comparisons. All 159 outputs equal up to phase.

Result 5. Against VOQC and QUESO, on their own benchmark

Nam gate set $\{H, X, R_z, \mathrm{CNOT}\}$, 26 circuits

tool	gates	change	CNOT
input	10,481		4,612
TKET	11,762	$+12.2\,\%$	4,489
Qiskit L3	10,391	$-0.9\,\%$	4,511
QUESO (5 min)	9,765	$-6.8\,\%$	4,465
ours	9,480	$-9.6\,\%$	4,426
VOQC	8,417	$-19.7\,\%$	4,314
VOQC, then ours	**7,833**	$-25.3\,\%$	**4,007**

Same input files as the published comparisons. All 159 outputs equal up to phase.

- **VOQC wins alone.** It merges R_z gates across the whole circuit, our windows see 3 qubits.
- **We find what it misses.** mod5_4: 35 gates vs. 56.
- **Together** best overall, ours improves VOQC on 21/26.

Takeaway. Same lesson as with BQSKit. A global pass plus exact local cleanup beats either alone.

Every output is verified, even beyond 25 qubits

- **Up to 24 qubits.** $\|(U - V)\psi\|$ on random states: 250/250 paper outputs, 720/720 suite outputs, 57 large outputs (max $1.4 \cdot 10^{-10}$).
- **Clifford, Clifford+T.** Exact arithmetic.
- **Beyond.** Statevector is out of reach, and MQT QCEC gave no answer on routed 23-qubit adders.

Every output is verified, even beyond 25 qubits

- **Up to 24 qubits.** $\|(U - V)\psi\|$ on random states: 250/250 paper outputs, 720/720 suite outputs, 57 large outputs (max $1.4 \cdot 10^{-10}$).
- **Clifford, Clifford+T.** Exact arithmetic.
- **Beyond.** Statevector is out of reach, and MQT QCEC gave no answer on routed 23-qubit adders.

Replay certificates

- The optimiser logs every step. An independent checker replays it with small matrices on a few wires.
- **171/176** large outputs certified, up to 544,922 steps, at most 6.8 s each.
- 5 more were rejected by an older checker limit and are being re-checked.

Takeaway. Checking a whole circuit becomes many tiny local checks, so cost grows with the steps, not the width.

Part V

Wrap-up

Where it stands and what comes next.

Limitations

- Noise models, no real-hardware runs yet.
- Windows of at most 3 qubits, single-threaded Python, hours at 100 qubits.
- VOQC beats us alone on the Nam set, BQSKit on 2q gates.
- Quartz not run, QUESO at 5 min, BQSKit on the smaller large circuits only.

Limitations and next steps

Limitations

- Noise models, no real-hardware runs yet.
- Windows of at most 3 qubits, single-threaded Python, hours at 100 qubits.
- VOQC beats us alone on the Nam set, BQSKit on 2q gates.
- Quartz not run, QUESO at 5 min, BQSKit on the smaller large circuits only.

Next

- Real IBM hardware. The driver is ready and tested on a fake backend, it needs an IBM account.
- Reach VOQC's rotation merging and BQSKit's 2q counts exactly (phase-polynomial passes, larger windows).
- Quartz baseline, QUESO with its full hour.

Summary

1. The original method is **3–4.5**$\times$ off the proven optimum, and the database radius is why.
2. A general optimiser with deep databases and convex windows reaches the optimum on **35/40** circuits without knowing the group.
3. On real algorithms it is a safe post-pass: after Qiskit L3 it never adds a 2q gate and removes 13–19 % of gates, and after BQSKit it removes 16 % more.
4. The gains are stable over seeds, and window length and database depth drive them.
5. Against VOQC and QUESO, ours is second alone and VOQC followed by ours is best.
6. Every output is verified, by simulation or by replay certificate.